CUHK Scientific Layout Families

Model, evidence, imaging, and next-experiment pages

Research Presentation Program

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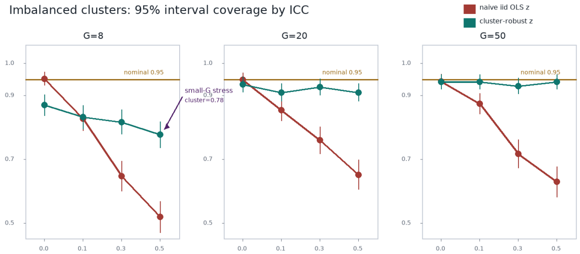
Clustered outcomes separate center and individual variation

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_j + \varepsilon_{ij}, \quad u_j \sim N(0, \tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

Center effects set $\rho = \tau^2/(\tau^2 + \sigma^2)$, so interval calibration depends on cluster count.

Cluster-robust intervals recover coverage as ICC increases

Nominal 0.95 remains visible; the stress point is the small-center setting.



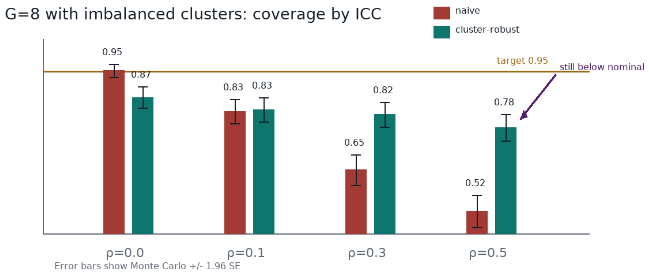
Synthetic clustered-data example; intervals compared on coverage, width, and bias.

Simulation varies clustering before interval procedures are compared

DGP knobs → Clustered samples → Interval procedures → Coverage diagnostics

The connector direction encodes data generation before interval estimation and coverage diagnostics.

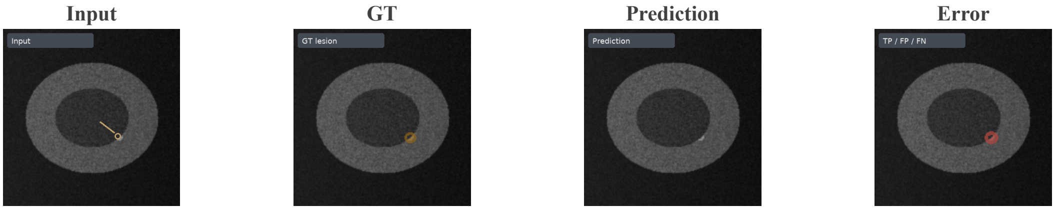
Small-G settings remain anti-conservative after robustification



Negative evidence is completed: CR2 and wild cluster bootstrap remain the next test.

The failure region is adjacent to the diagnostic explanation instead of hidden in a note.

Same-case panels keep the segmentation error interpretable



All four panels come from the same case; the error panel binds the failure explanation.

Next experiment tests whether batch selection reduces fragile coverage

Small-G limit → DPP batch query → Mondrian partition → CR2 / wild bootstrap

Coverage evidence determines which batch-query strategy to validate next; CR2 and wild bootstrap remain proposed comparators.